

UML Architecture Package

Physiological Digital Twin Berbasis CAPAR

Dokumen ini mendefinisikan representasi UML tingkat arsitektur untuk sistem Physiological Digital Twin CAPAR. Package terdiri dari Use Case Diagram, Component Diagram, Sequence Diagram, dan Deployment Diagram dalam bentuk spesifikasi tekstual yang dapat diterjemahkan menjadi model UML.

1. Use Case Diagram

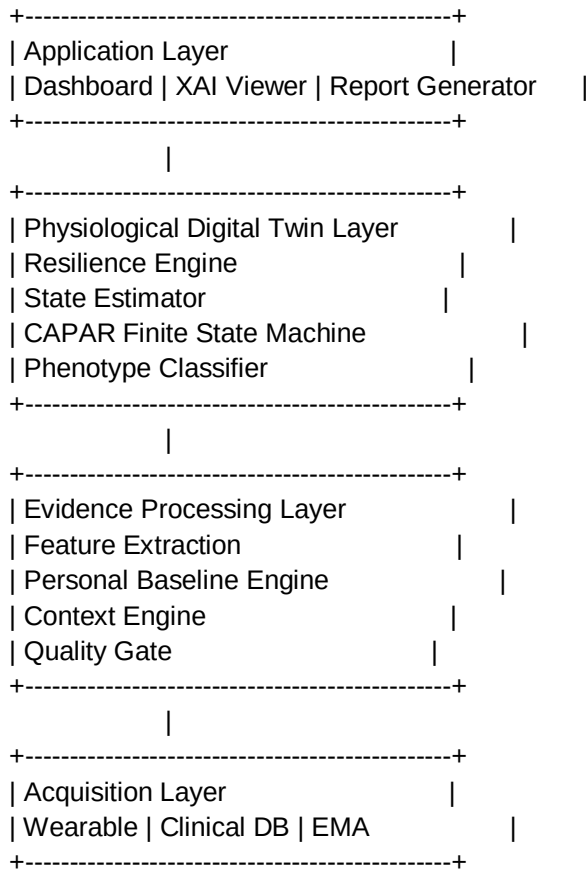
Actors:

1. Participant
 - memakai wearable
 - memberikan EMA/context
 - melihat hasil personal
2. Clinician/Researcher
 - melihat resilience status
 - mengevaluasi trajectory
 - membaca explanation
3. Concept Engineer
 - mengatur physiological model
 - mengatur rule dan threshold
4. Software Engineer
 - melakukan deployment
 - maintenance sistem

Main Use Cases:

- UC-01 Acquire Physiological Data
- UC-02 Manage Context Input
- UC-03 Integrate Clinical Information
- UC-04 Process Physiological Signal
- UC-05 Extract CAPAR Features
- UC-06 Estimate Resilience State
- UC-07 Generate CAPAR Event-State
- UC-08 Display Recovery Trajectory
- UC-09 Generate XAI Explanation
- UC-10 Export Research Report

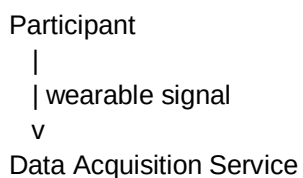
2. Component Diagram

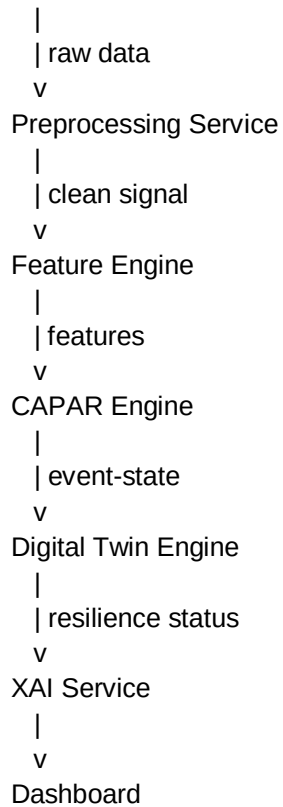


3. Component Responsibility

- Data Acquisition Component: menerima data sensor dan clinical input.
- Quality Component: melakukan validasi dan confidence scoring.
- Feature Component: menghitung HR, HRV, DFA, recovery, relapse feature.
- Baseline Component: membuat baseline personal-contextual.
- CAPAR Engine: melakukan deviation, persistence, recovery, relapse detection.
- Digital Twin Engine: memperkirakan resilience state.
- XAI Engine: menyediakan evidence dan uncertainty.

4. Sequence Diagram: Continuous Monitoring





5. Sequence Logic CAPAR Engine

Input:

$u(k)$, $d(k)$, $c(k)$, $y(k)$

Process:

1. Validate observation.
2. Determine context.
3. Compare with personal baseline.
4. Calculate deviation score.
5. Evaluate persistence.
6. Detect recovery.
7. Detect relapse.
8. Update state.

Output:

$s(k)$:

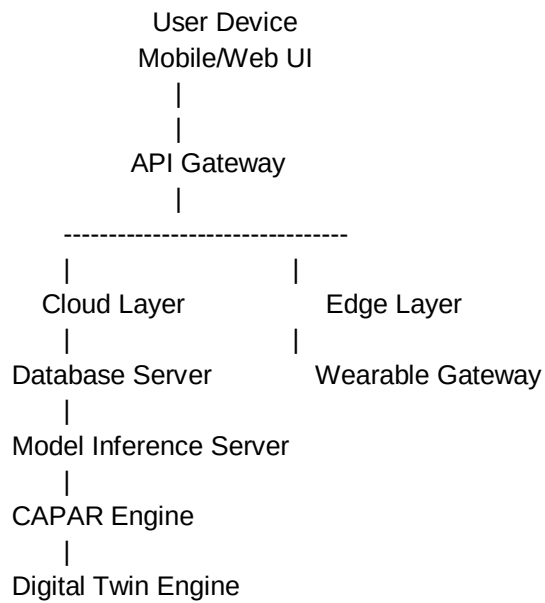
Baseline Mature

No Deviation

Deviation

Persistent Deviation
Recovery Start
Full Recovery
Relapse

6. Deployment Diagram



7. Traceability Architecture

Input:
Wearable + Clinical + Context



Feature:

HR, HRV, DFA, recovery metrics



Latent Variable:

Autonomic tone
Cardiac reserve
Recovery dynamics



Resilience Domain:

Vulnerability
Cardiac Reserve
Autonomic Reserve
Recovery Capacity
Regulation Stability



Output:

Risk
Trajectory
Phenotype
Decision Support
XAI

8. Architectural Validation

Validasi UML dilakukan dengan memastikan setiap requirement SRS memiliki representasi pada komponen arsitektur dan setiap komponen memiliki tanggung jawab yang jelas.